LAB 9: Numerical Integration

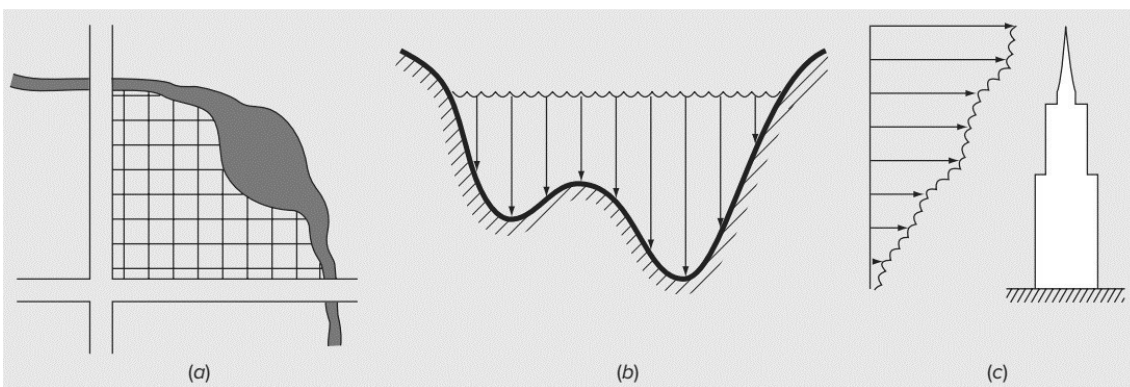
Objective: To numerically obtain the integration of any given function.

Theory: Mathematically, definite integration is represented by

$$I = \int_a^b f(x)\,dx$$

which stands for the integral of the function $f(x)$ with respect to the independent variable x, evaluated between the limits $x = a$ to $x = b$.

The meaning of integration is the total value, or summation, of a f (x) dx over the range x = a to b. In fact, the symbol ∫ is actually a stylized capital S that is intended to signify the close connection between integration and summation.



Examples of how integration is used to evaluate areas in engineering and scientific applications. (*a*) A surveyor might need to know the area of a field bounded by a meandering stream and two roads. (*b*) A hydrologist might need to know the cross-sectional area of a river. (*c*) A structural engineer might need to determine the net force due to a nonuniform wind blowing against the side of a skyscraper.

Method to numerically evaluate integration-

TRAPEZOIDAL RULE –

The *Newton-Cotes formulas* are the most common numerical integration schemes. They are based on the strategy of replacing a complicated function or tabulated data with a polynomial that is easy to integrate:

$$I = \int_a^b f(x)\,dx \cong \int_a^b f_n(x)\,dx$$

where $f_n(x)$ = a polynomial of the form

$$f_n(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} + a_n x^n$$

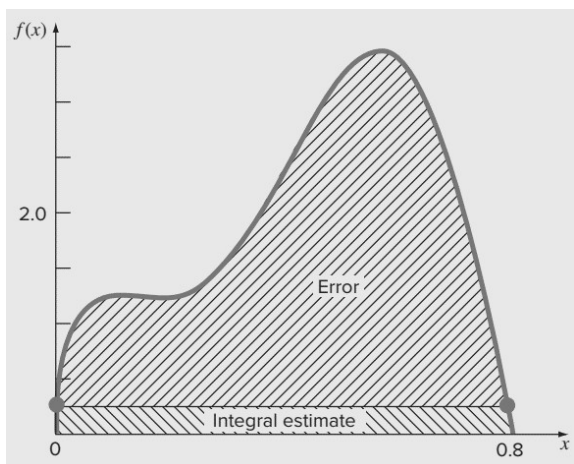
The trapezoidal rule is the first of the Newton-Cotes closed integration formulas as linear assumption of polynomial i.e. straight line function as, $f(x) = a_0+a_1x$. Then the integral I is calculated between a and b as,

$$I = \int_a^b \left[f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right] dx$$

The result of the integration is

$$I = (b - a)\frac{f(a) + f(b)}{2} \qquad I = \text{width} \times \text{average height}$$

which is called the *trapezoidal rule*.

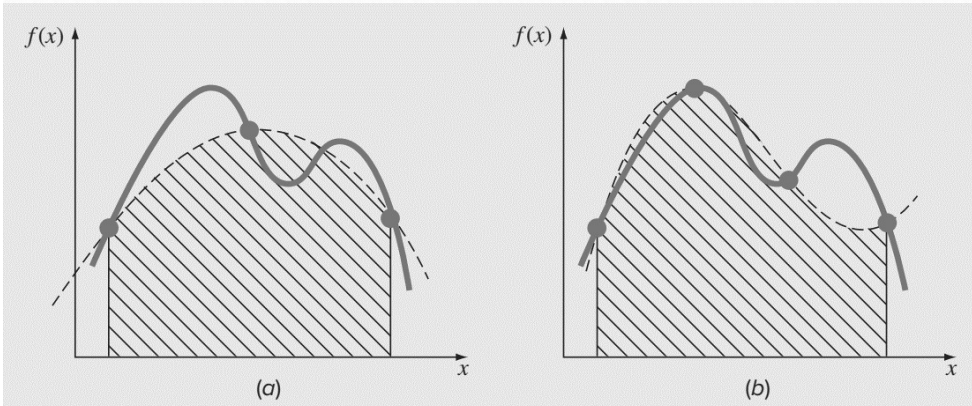


Error estimate

Simpson's 1/3 Rule –

Simpson's 1/3 rule corresponds to the case where the polynomial higher degree i.e. parabola is assumed using 3 points.

(*a*) Graphical depiction of Simpson's 1/3 rule: It consists of taking the area under a parabola connecting three points. (*b*) Graphical depiction of Simpson's 3/8 rule: It consists of taking the area under a cubic equation connecting four points.



$$I = \int_{x_0}^{x_2} \left[\frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) \right.$$

$$\left. + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2) \right] dx$$

$$I = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

where, for this case, $h = (b - a)/2$. This equation is known as *Simpson's 1/3 rule*.

$$I = (b - a) \frac{f(x_0) + 4f(x_1) + f(x_2)}{6}$$

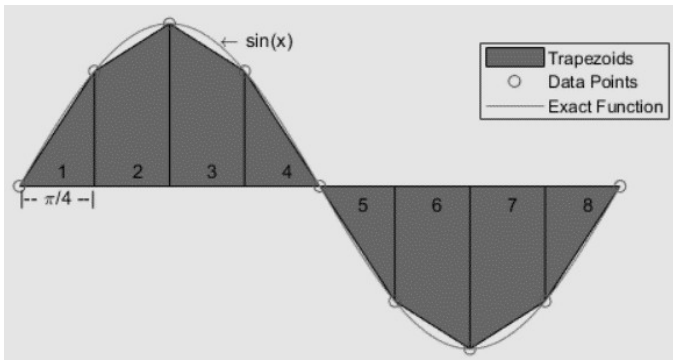
where $a = x_0$, $b = x_2$, and x_1 = the point midway between a and b, which is given by $(a + b)/2$.

Problem Statement. Use Eq. (19.23) to integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

from $a = 0$ to $b = 0.8$ using analytical, Trapezoidal and 1/3 simpson rule.

Segments (n)	Points	Name	Formula	Truncation Error
1	2	Trapezoidal rule	$(b-a)\dfrac{f(x_0)+f(x_1)}{2}$	$-(1/12)h^3 f''(\xi)$
2	3	Simpson's 1/3 rule	$(b-a)\dfrac{f(x_0)+4f(x_1)+f(x_2)}{6}$	$-(1/90)h^5 f^{(4)}(\xi)$
3	4	Simpson's 3/8 rule	$(b-a)\dfrac{f(x_0)+3f(x_1)+3f(x_2)+f(x_3)}{8}$	$-(3/80)\, h^5 f^{(4)}(\xi)$



`trapz` performs numerical integration via the trapezoidal method. This method approximates the integration over an interval by breaking the area down into trapezoids with more easily computable areas. For example, here is a trapezoidal integration of the sine function using eight evenly-spaced trapezoids:

For an integration with `N+1` evenly spaced points, the approximation is

$$\int_a^b f(x)dx \approx \frac{b-a}{2N}\sum_{n=1}^{N}(f(x_n)+f(x_{n+1}))$$

$$= \frac{b-a}{2N}[f(x_1)+2f(x_2)+\ldots+2f(x_N)+f(x_{N+1})] \ ,$$

where the spacing between each point is equal to the scalar value $\frac{b-a}{N}$. By default MATLAB® uses a spacing of 1.

If the spacing between the `N+1` points is not constant, then the formula generalizes to

$$\int_a^b f(x)dx \approx \frac{1}{2}\sum_{n=1}^{N}(x_{n+1}-x_n)[f(x_n)+f(x_{n+1})] \ ,$$

where $a = x_1 < x_2 < \ \ldots \ < x_N < x_{N+1} = b$, and $(x_{n+1}-x_n)$ is the spacing between each consecutive pair of points.

MATLAB code for evaluating area under the sin curve.

```
clc;clear all; close all;
X = 0:pi/100:pi;
Y = sin(X);
Q = trapz(X,Y);
```

Assignment -

One numerical as suggested by Faculty.

Sample numerical.

19.4 Evaluate the following integral:

$$\int_{-2}^{4} (1 - x - 4x^3 + 2x^5)\, dx$$

(a) analytically, **(b)** single application of the trapezoidal rule, **(c)** composite trapezoidal rule with $n = 2$ and 4, **(d)** single application of Simpson's 1/3 rule, **(e)** Simpson's 3/8 rule, and

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Experiment - 9.

Numerical :-

a) Analytical :-

$$\int_{-2}^{4} (1 - x - 4x^3 + 2x^5)\,dx$$

$$\Rightarrow \left[x - \frac{x^2}{2} - x^4 + \frac{x^6}{3} \right]_{-2}^{4} \quad \text{— (1)}$$

b) Trapezoidal rule (single integral)

$$I = \frac{b-a}{2} \left(f(a) - f(b) \right) \qquad \begin{bmatrix} a = -2 \\ b = 4 \end{bmatrix}$$

$$f(-2) = -2$$
$$f(4) = 1664$$

c) Composite trapezoidal rule with $n = 2$ & 4,

$$I = \frac{h}{2} \left(f_0 + 2f_1 + f_2 \right)$$

d) Trapezoidal rule $(n = 4)$

$$I = \frac{h}{2} \left(f_0 + 2(f_1 + f_2 + f_3) + f_4 \right)$$

e) Simpson's $\left(\frac{1}{3}\right)^{rd}$ rule :-

$$I = \frac{h}{3} \left(f_0 + 4f_1 + f_2 \right)$$

(f) Simson's $\left(\frac{3}{8}\right)^{th}$ rule :-

$$I = \frac{3h}{8} \left(f_0 + 3f_1 + 3f_2 + f_3 \right)$$

```matlab
a = -2;
b = 4;

%Analytical

% F(x) = x - x^2/2 - x^4 + x^6/3

Fa = a - (a^2)/2 - (a^4) + (a^6)/3;
Fb = b - (b^2)/2 - (b^4) + (b^6)/3;

I1 = Fb - Fa;

%trapazoidal

fa = 1 - a - 4*a^3 + 2*a^5;
fb = 1 - b - 4*b^3 + 2*b^5;

I2 = ((b - a)/2) * (fa + fb);

% composite trapazoidal

n_values = [2 4];

for k = 1:length(n_values)

    n = n_values(k);
    h = (b - a)/n;

    sum = 0;

    for i = 1:n-1
        xi = a + i*h;
        fi = 1 - xi - 4*xi^3 + 2*xi^5;
        sum = sum + fi;
    end

    f0 = 1 - a - 4*a^3 + 2*a^5;
    fn = 1 - b - 4*b^3 + 2*b^5;

    I_trap(k) = (h/2) * (f0 + 2*sum + fn);
end

I3 = I_trap(1); % n = 2
I4 = I_trap(2); % n = 4

%simpson 1/3

h = (b - a)/2;
x1 = (a + b)/2;

f0 = 1 - a - 4*a^3 + 2*a^5;
f1 = 1 - x1 - 4*x1^3 + 2*x1^5;
f2 = 1 - b - 4*b^3 + 2*b^5;

I5 = (h/3) * (f0 + 4*f1 + f2);

%simpson 3/8

n = 3;
h = (b - a)/n;

sum = 0;

for i = 1:n-1
    xi = a + i*h;
    fi = 1 - xi - 4*xi^3 + 2*xi^5;
    sum = sum + 3*fi;
end

f0 = 1 - a - 4*a^3 + 2*a^5;
fn = 1 - b - 4*b^3 + 2*b^5;

I6 = (3*h/8) * (f0 + sum + fn);

fprintf('Analytical   = %.2f\n', I1);
fprintf('Trap single  = %.2f\n', I2);
fprintf('Trap n=2     = %.2f\n', I3);
fprintf('Trap n=4     = %.2f\n', I4);
fprintf('Simpson 1/3  = %.2f\n', I5);
fprintf('Simpson 3/8  = %.2f\n', I6);
```

Figure 9.1 MATLAB Code

```
Command Window
>> CEL_EXP_9
Analytical    = 1104.00
Trap single   = 5280.00
Trap n=2      = 2634.00
Trap n=4      = 1516.88
Simpson 1/3   = 1752.00
Simpson 3/8   = 1392.00
>>
```

Figure 9.2 MATLAB Result

Conclusion

The analytical solution of the integral is 1104, which serves as the exact reference value. All numerical methods produced approximate results with varying accuracy.

The single trapezoidal rule showed a very large deviation (5280, error ≈ 378%), indicating poor accuracy for functions with high curvature over large intervals. The composite trapezoidal rule improved the result significantly:

- For n = 2, result = 2634 (error ≈ 138.6%)
- For n = 4, result = 1516.88 (error ≈ 37.4%)

This confirms that increasing the number of intervals reduces error.
Among all methods, Simpson's rules performed better due to higher-order interpolation:

- Simpson's 1/3 rule: 1752 (error ≈ 58.7%)
- Simpson's 3/8 rule: 1392 (error ≈ 26.1%)

The Simpson's 3/8 rule gave the closest result to the analytical value among the tested methods.

Overall, the experiment demonstrates that:

- Accuracy improves with higher-order methods (Simpson > Trapezoidal)
- Accuracy increases with more subintervals
- Numerical methods converge toward the analytical solution but may require finer discretization for highly nonlinear functions